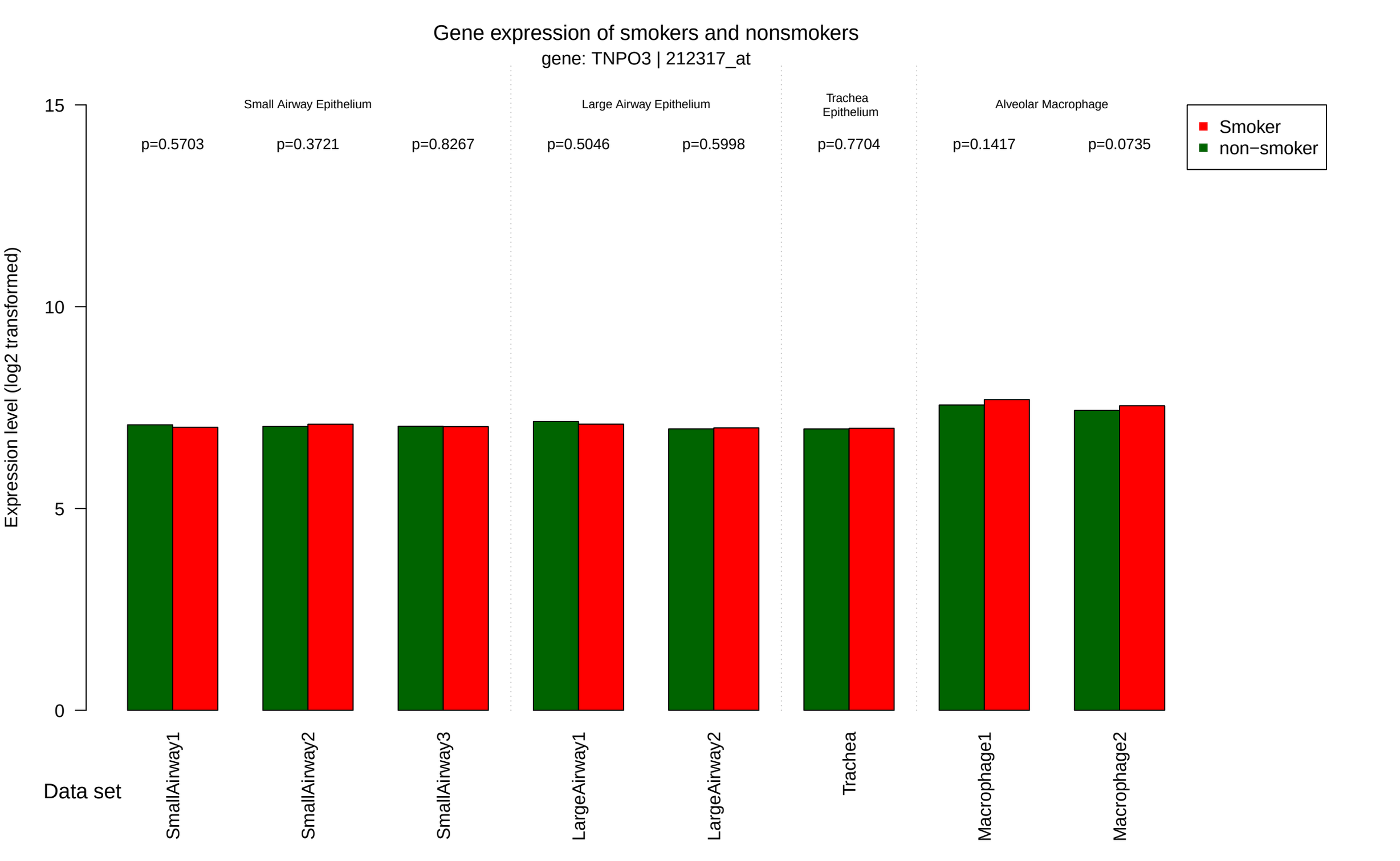
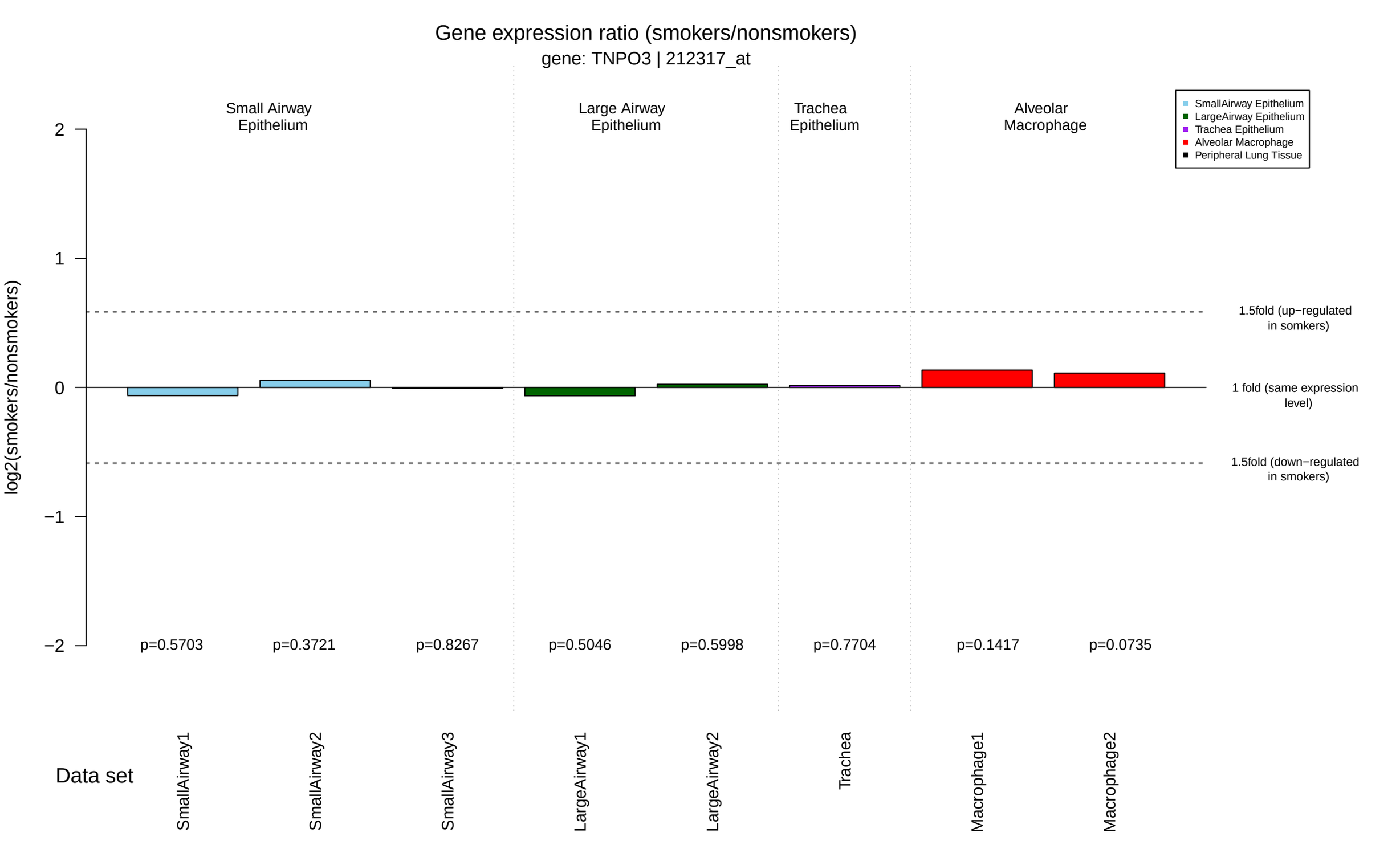
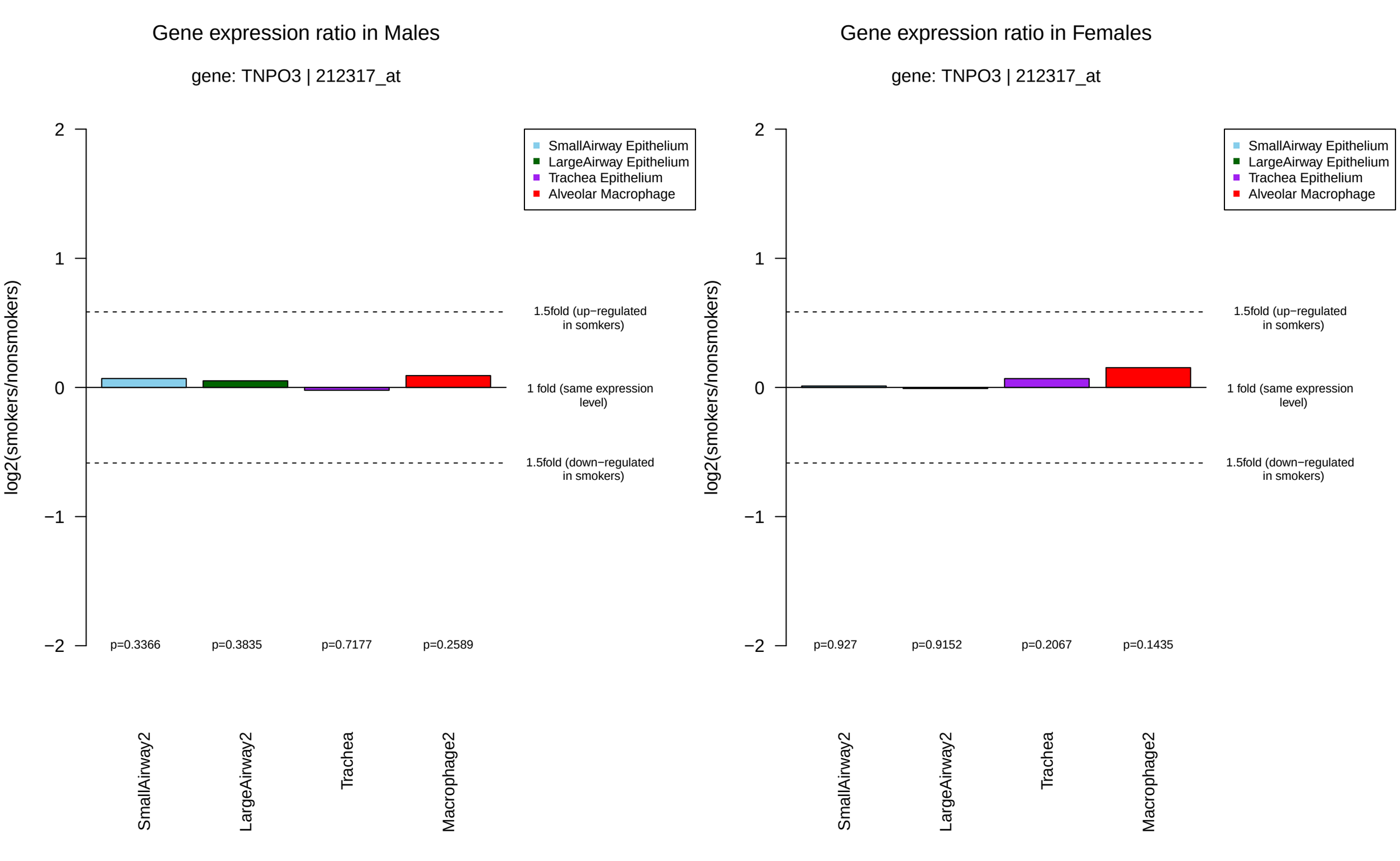
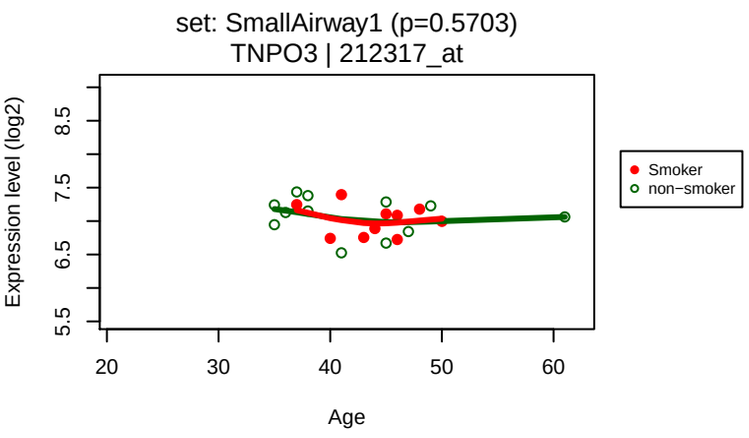
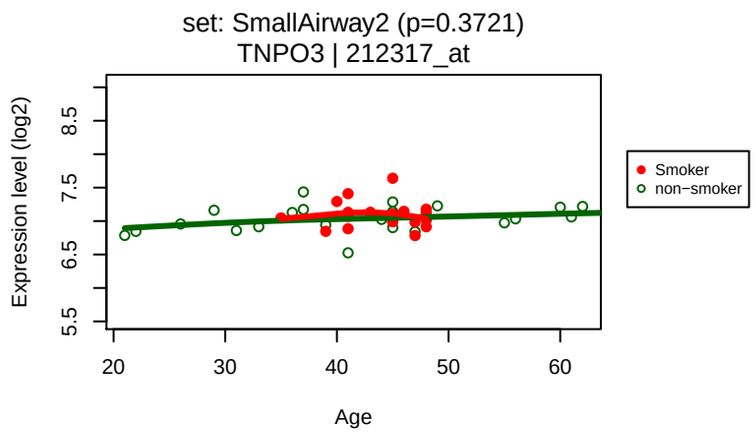
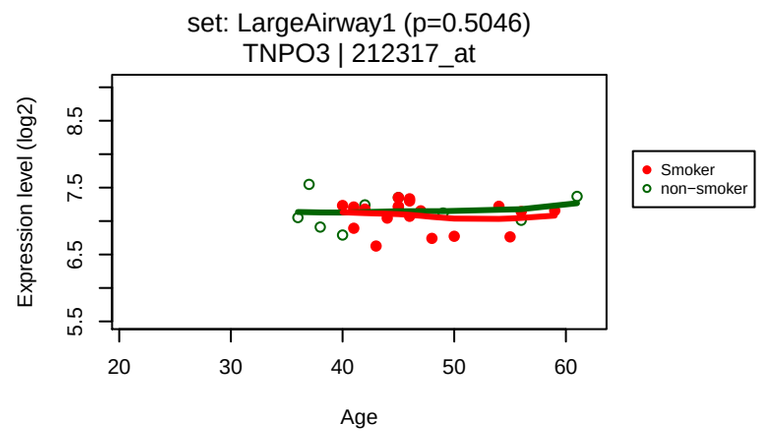

Gene expression of smokers and nonsmokers
gene: TNPO3 | 212317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5703
p=0.3721
p=0.8267
p=0.5046
p=0.5998
p=0.7704
p=0.1417
p=0.0735
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

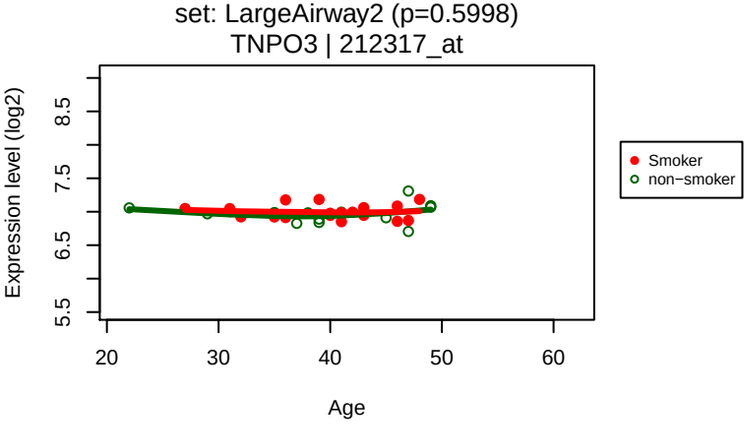

Gene expression ratio (smokers/nonsmokers)
gene: TNPO3 | 212317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5703
p=0.3721
p=0.8267
p=0.5046
p=0.5998
p=0.7704
p=0.1417
p=0.0735
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

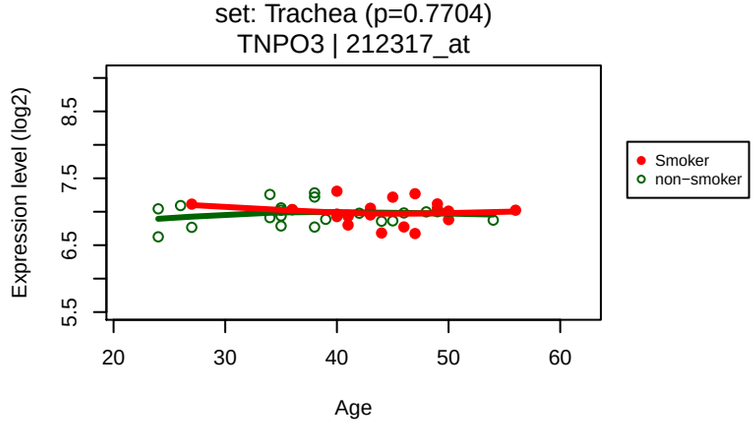

Gene expression ratio in Males
gene: TNPO3 | 212317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3366
p=0.3835
p=0.7177
p=0.2589
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TNPO3 | 212317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.927
p=0.9152
p=0.2067
p=0.1435
SmallAirway2
LargeAirway2
Trachea
Macrophage2

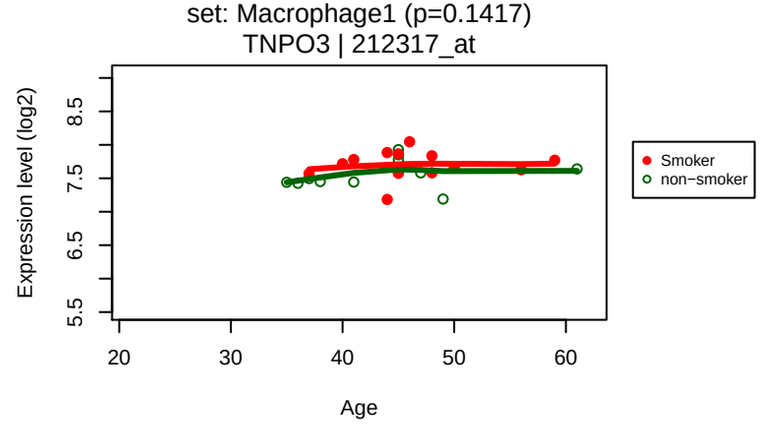

set: SmallAirway1 (p=0.5703)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

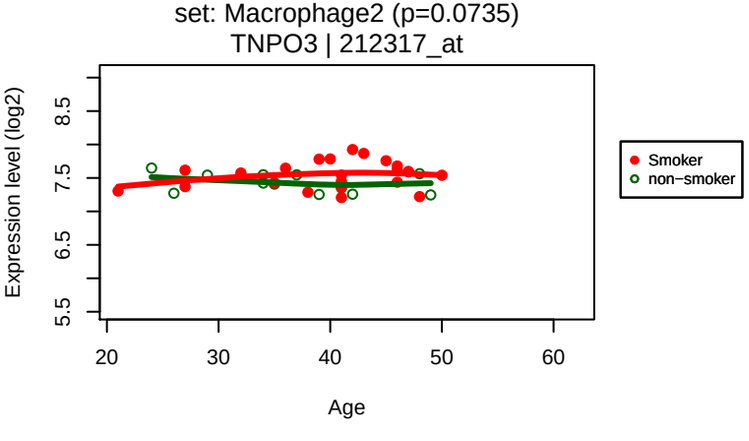

set: SmallAirway2 (p=0.3721)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

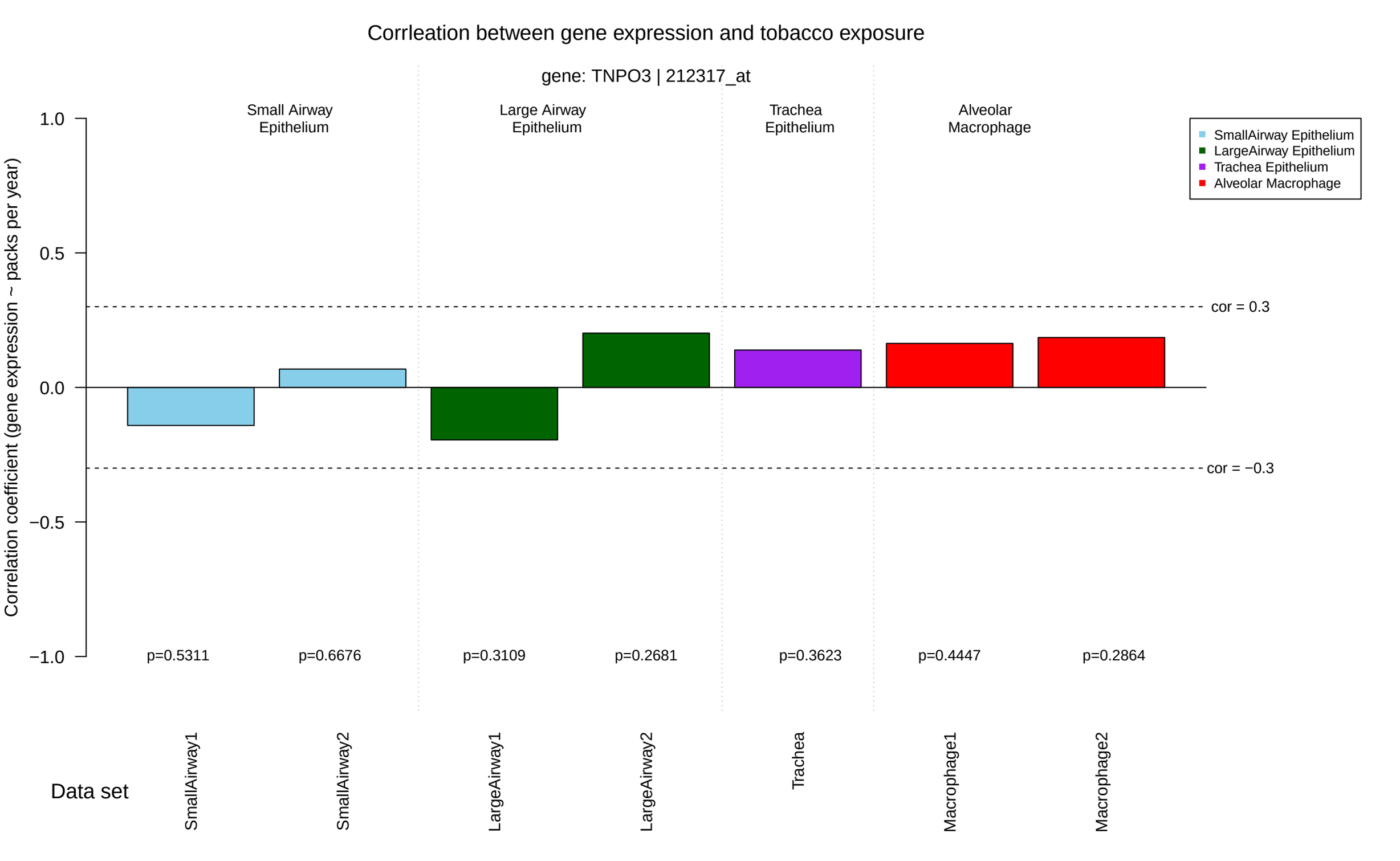

set: LargeAirway1 (p=0.5046)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5998)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7704)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1417)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0735)
TNPO3 | 212317_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TNPO3 | 212317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5311
p=0.6676
p=0.3109
p=0.2681
p=0.3623
p=0.4447
p=0.2864
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2